

Digital Forensics Laboratory

EXPERIMENT NO: 04

DATE: August 6, 2026

TITLE: Analyze Email Headers and Detect Email Spoofing using MHA (Mail Header Analyzer)

1. AIM

To extract, analyze, and interpret email headers to trace an email's origin, verify its authenticity protocols (SPF, DKIM, DMARC), and detect potential spoofing or phishing attempts using Mail Header Analyzer (MHA) tools.

2. SOFTWARE / TOOLS REQUIRED

- **Operating System:** Windows, Linux, or macOS
- **Email Client:** Web browser accessing Gmail, Outlook, or Yahoo
- **Analysis Tools:** MXToolbox (Online Mail Header Analyzer), WHOIS IP Lookup

3. THEORY

Every email sent over the internet contains hidden metadata known as an **Email Header**. This header tracks the email's journey from the sender's server to the recipient's inbox and contains critical routing and authentication information. Analyzing these headers helps forensic investigators verify if an email is legitimate or spoofed.

Key Authentication Protocols:

- **SPF (Sender Policy Framework):** Checks if the sender's IP address is authorized to send emails on behalf of the claimed domain.
- **DKIM (DomainKeys Identified Mail):** A cryptographic signature that verifies the email content has not been altered in transit.
- **DMARC (Domain-based Message Authentication, Reporting, and Conformance):** A policy that uses both SPF and DKIM to authenticate the email and instruct the receiving server on how to handle failures.

4. PROCEDURE & OBSERVATIONS

Step 1: Accessing the Suspicious Email

- Log into your email account (e.g., Gmail) and locate the target email you wish to investigate (often found in the Spam or Inbox folder).
- Open the email to view the sender details and subject line.

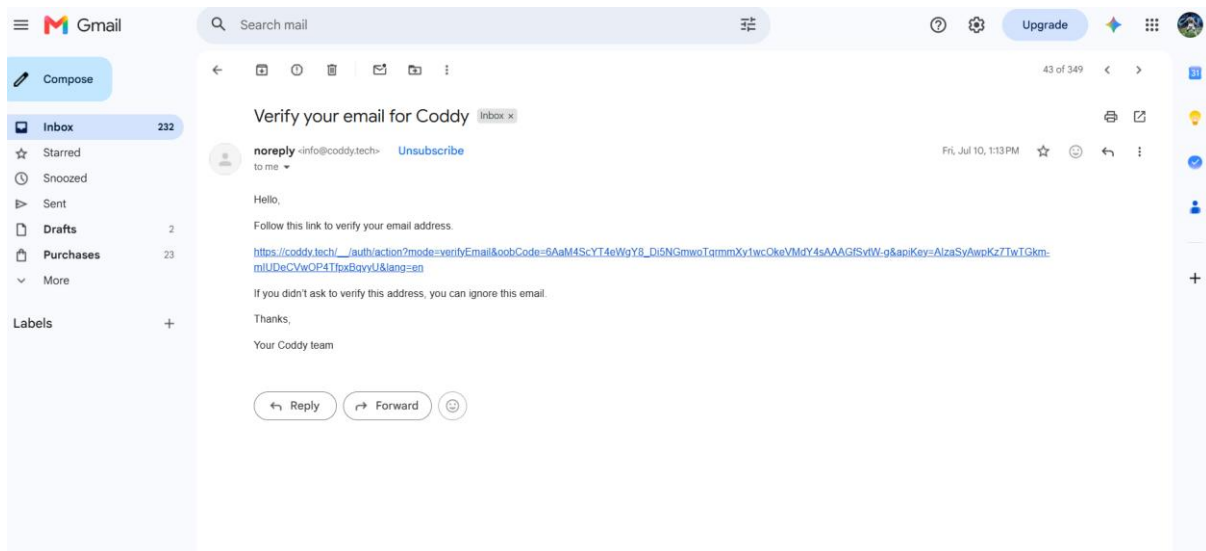


Figure 1: Viewing the suspicious email in the Gmail Spam folder

Step 2: Extracting the Raw Email Header

1. To view the hidden metadata in Gmail, click the three dots (More) in the upper right corner of the email and select **"Show original"**. (For Outlook: File > Properties > Internet headers. For Yahoo: More > View raw message).
2. Copy all the raw text provided. This text contains the Received fields, Message-ID, and authentication results.

Original Message

Message ID	<0bee3fe9-86e6-412c-ba57-6223484653d3@smtp-relay.sendinblue.com>
Created at:	Fri, Jul 10, 2026 at 1:13 PM (Delivered after 2 seconds)
From:	noreply <info@coddy.tech>
To:	jovanmariyawilliam@gmail.com
Subject:	Verify your email for Coddy
SPF:	PASS with IP 77.32.148.23 Learn more
DKIM:	'PASS' with domain coddy.tech Learn more
DMARC:	'PASS' Learn more

[Download Original](#)
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Delivered-To: jovanmariyawilliam@gmail.com

Received: by 2002:ab3:7796:0:b0:307:dfb:3332 with SMTP id n22csp1057163ltj; Fri, 10 Jul 2026 00:43:36 -0700 (PDT)

X-Received: by 2002:ac2:4bd0:0:b0:5b0:1879:1ba5 with SMTP id 2adb3069b0e04-5b018791d32mr1243269e87.58.1783669416001; Fri, 10 Jul 2026 00:43:36 -0700 (PDT)

ARC-Seal: i=1; a=rsa-sha256; t=1783669415; cv=none; d=google.com; s=arc-20260327; b=pk8wHPGM5PP7dperQCSgVbpaCn9rwe2Mrr+RcWlpw3Y1WqI4SwYPpd4TGfytF4VnaKFuy151CcbhL4Quvrh7RQJ1xZ1th+s7f7wJwqAYkUG5svXdf+4H0dMXGMQ1uK2yOS KmvJMS5Be1HD12FDXohvNBh/30Xrat7D6J7jUExdI19bF9UkRs5itiG80r7FRztQRHEZ49uNLvinF3EQ9VnrokkcD7TPimgHKFFCLBv9U2w+4agBHV1FHYt2gfQdJ+r6zo1z1Dl WDM0djr2yCJWf2ti91hjoX0U0V99ebXzRKjni6kh29ruPytge9ySnShdjc3JcHqIM RMUQ==

ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=mime-version:list-unsubscribe:from:subject:feedback-id:list-unsubscribe-post:origin-messageid:message-id:date:to:dkim-signature; bh=ismfMYIj3mB/3DtJ3QezCwxdBWH1gfSaMLSRfdUUU=; fh=7uZPOe0z6TD09HutbGfHfpu3/17sZ6sF0xx5WfKujw=; b=akoliotJmno9ZCjAU1XZdanKk1q8kd26Px1h48Wljawyd00a380ecj1/ppf89vpAW0z4q1Z1AmcAycp7vxeCEK0BilGb6Bw6EFyUpkNhy9NHtFLXBpMXLEJd3j00SAyJpW09 cGh08+TicW4Z1JuoYB6FGya9pEeQUYJgBRsgfzpk9wb240/Q3iCTo3leBwt+Ado6617f5eahDAmf2M5GmQrZ1iEF02YC0fiq1ax5bkunsHDXvcDY0M0Mm8mMo2VxJYNxfQj2 cSfwjwgIP8I7UuoC4GdeUGxdy1tr21IbAg9ZU9mswj2k7kphUCrrao6RHSUBm1U7FVG doUw==;

dara=google.com

ARC-Authentication-Results: i=1; mx.google.com; dkim=pass header.i=coddy.tech header.s=mail header.b=U47V2kMp; spf=pass (google.com: domain of bounces-171700654-2707556336@gw.d.sender-sib.com designates 77.32.148.23 as permitted sender) smtp.mailfrom=bounces-171700654-2707556336@gw.d.sender-sib.com; dmarc=pass (p=NONE sp=NONE dis=NONE) header.from=coddy.tech

Return-Path: <bounces-171700654-2707556336@gw.d.sender-sib.com>

Received: from gw.d.sender-sib.com (gw.d.sender-sib.com. [77.32.148.23]) by mx.google.com with ESMTPS id 2adb3069b0e04-5b01ca9c742si456285e87.136.2026.07.10.00.43.35 for <jovanmariyawilliam@gmail.com> (version=TLS3_3 cipher=TLS_AES_256_GCM_SHA384 bits=256/256); Fri, 10 Jul 2026 00:43:35 -0700 (PDT)

Figure 2: Extracting the original raw message and email header data

Step 3: Analyzing the Header via MXToolbox

1. Navigate to an online Mail Header Analyzer, such as **MXToolbox**.

2. Paste the copied raw header text into the analysis tool and click Analyze.
3. Review the **Delivery Information** and **Relay Information** to trace the chronological path (hops) the email took from the sending server to the receiving server. Check for unusual delays or suspicious routing.

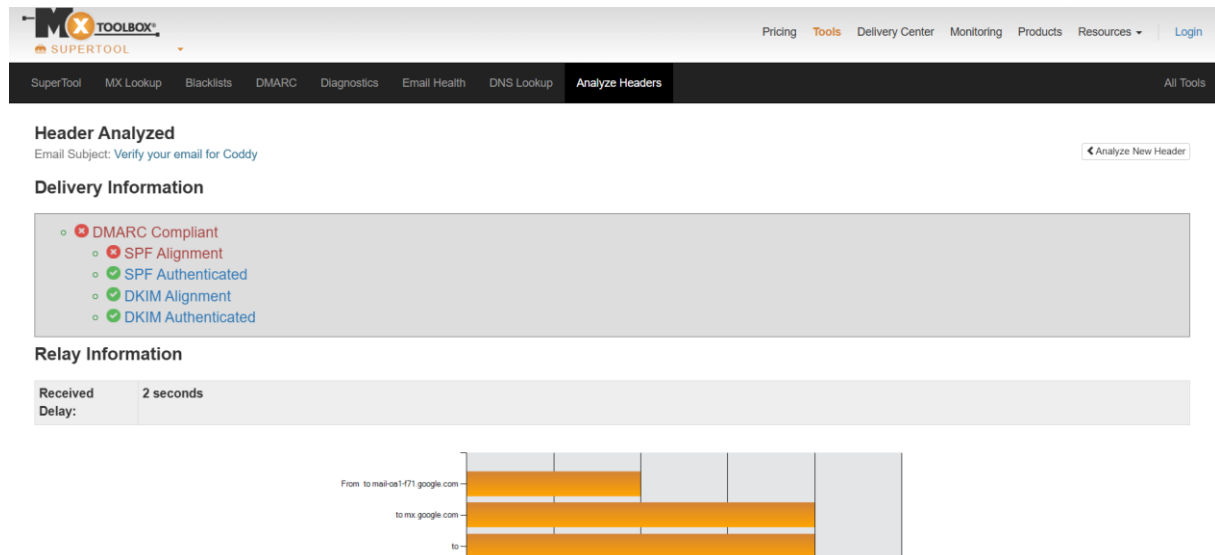


Figure 3: MHA relay analysis showing chronological hops and delivery delays

Step 4: Examining SPF, DKIM, and DMARC Results

1. Scroll down in the analyzer tool to inspect the authentication policies and Verify if the DMARC record is compliant.
2. Check the SPF and DKIM alignment to ensure the sender's IP (185.255.8.37) was genuinely authorized by the domain (communication.vodafoneidea.in).

Hop	Delay	From	By	With	Time (UTC)	Blacklist
1	*		mail-0a1-f71.google.com	SMTP	7/10/2026 7:43:34 AM	
2	1 Second	gw.d.sender-sib.com 77.32.148.23	mx.google.com	ESMTPS	7/10/2026 7:43:35 AM	✖
3	1 Second		2002.ab3.7796.0.b0.307.dfb.3332	SMTP	7/10/2026 7:43:36 AM	

EmailHeaders advertisement image

SPF and DKIM Information

dmarc:coddy.tech Hide Solve Email Delivery Problems			
v=DMARC1; p=none; rua=mailto:info@coddy.tech;			
Tag	TagValue	Name	Description
v	DMARC1	Version	Identifies the record retrieved as a DMARC record. It must be the first tag in the list.
p	none	Policy	Policy to apply to email that fails the DMARC test. Valid values can be 'none', 'quarantine', or 'reject'.
rua	mailto:info@coddy.tech	Receivers	Addresses to which aggregate feedback is to be sent. Comma separated plain-text list of DMARC URIs.
Test		Result	
✖	DMARC Policy Not Enabled	DMARC Quarantine/Reject policy not enabled	
✔	DMARC Record Published	DMARC Record found	
✔	DMARC Syntax Check	The record is valid	
✔	DMARC Multiple Records	Multiple DMARC records corrected to a single record.	

Figure 4: Verification of SPF, DKIM, and DMARC cryptographic records

Step 5: IP Address and Hostname Verification

1. Extract the primary sending IP address identified in the Received lines (185.255.8.37).

2. Use a **WHOIS** lookup tool to identify the geographical location and ownership of the IP address.
3. Determine if the registered organization matches the expected sender.

The screenshot shows a web interface for a Whois IP lookup tool. The header includes the 'Whois' logo and navigation links for Domains, Hosting, Websites, Email, Security, Whois, and Deals. A search bar on the right contains the IP address '193.35.16.213'. The main heading is 'Whois IP 193.35.16.213', with a timestamp 'Updated 1 second ago' to its right. Below the heading is a large box containing the RIPE Database query results in RPSL format. The results include a note about the output being filtered, information related to the IP range '193.35.16.0 - 193.35.16.255', and an abuse contact 'abuse@netcore.co.in'. A detailed list of object attributes follows, including inetnum, mnt-by, netname, descr, country, admin-c, abuse-c, tech-c, org, status, mnt-lower, mnt-domains, mnt-routes, created, last-modified, and source.

```
% This is the RIPE Database query service.
% The objects are in RPSL format.
%
% The RIPE Database is subject to Terms and Conditions.
% See https://docs.db.ripe.net/terms-conditions.html

% Note: this output has been filtered.
%       To receive output for a database update, use the "-B" flag.

% Information related to '193.35.16.0 - 193.35.16.255'

% Abuse contact for '193.35.16.0 - 193.35.16.255' is 'abuse@netcore.co.in'

inetnum:        193.35.16.0 - 193.35.16.255
mnt-by:         MNT-NETERRA
netname:        Netcore_Cloud-NET
descr:          Netcore_Cloud
country:        US
admin-c:        FS16232-RIPE
abuse-c:        AR65331-RIPE
descr:          abuse@netcore.co.in
tech-c:         FS16232-RIPE
org:            ORG-FA1221-RIPE
status:         ASSIGNED PA
mnt-lower:      FEEDHOSTING-MNT
mnt-domains:    FEEDHOSTING-MNT
mnt-routes:     FEEDHOSTING-MNT
created:        2022-05-30T11:57:15Z
last-modified:  2025-04-28T12:35:58Z
source:         RIPE
```

Figure 5: WHOIS IP lookup revealing the hosting organization (INFOBIP_INDIA)

5.FORENSICS FINDINGS AND ANALYSIS

Based on the header analysis conducted:

- **Sender Domain:** communication.vodafoneidea.in
- **Sending IP Address:** 193.35.16.213
- **Authentication Status:** The email **PASSED** all primary checks (SPF, DKIM, and DMARC). The IP address is officially authorized to send on behalf of the domain.

- **WHOIS Verification:** The IP belongs to INFOBIP_INDIA, a legitimate bulk SMS and email gateway service frequently used by telecom providers.
- **Conclusion on Spoofing:** Despite being flagged as spam by Gmail's automated content filters (likely due to promotional marketing content), the header analysis proves this email is **cryptographically authentic and not spoofed**. The sender identity is genuine.

6.RESULT

The email header was successfully extracted and analyzed using MXToolbox. Authentication protocols (SPF, DKIM, DMARC) and the originating IP address were manually verified, demonstrating how to trace email origins and detect potential spoofing.